ETSI TS 128 556 V18.0.0 (2024-05)



5G;
Management and orchestration;
Network policy management for 5G mobile networks;
Stage 2 and stage 3
(3GPP TS 28.556 version 18.0.0 Release 18)



Reference
RTS/TSGS-0528556vi00

Keywords
5G

ETSI

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Contents

Intellectual Property Rights 2
Legal Notice 2
Modal verbs terminology 2
Foreword 5
Introduction 6
1 Scope 7
2 References 7
3 Definitions of terms, symbols and abbreviations 7
3.1 Terms 7
3.2 Symbols 7
3.3 Abbreviations 7
4 Policy management procedures 8
4.1 Policy Creation 8
4.2 Policy Deletion 8
4.3 Policy Update 8
4.4 Policy Query 9
4.5 Policy Conflicts Notification 9
4.6 Policy Activation 9
4.7 Policy Deactivation 10
5 Policy MnS – Stage 2 10
5.1 Management operation for Policy (MnS component type A) 10
5.2 Information model definition entities for Policy 10
5.2.1 Imported information entities and local labels 10
5.2.2 Class diagram 11
5.2.2.1 Relationships 11
5.2.2.2 Inheritance 11
5.2.3 Class definitions 11
5.2.3.1 Policy 11
5.2.3.1.1 Definition 11
5.2.3.1.2 Attributes 11
5.2.3.2 Notifications 12
5.2.3.2.0 General 12
5.2.3.2.1 Notification `notifyPolicyConflict` (CM) 12
5.2.3.2.1.2 Notification information 12
5.2.3.3 PolicyContent <<dataType>> 12
5.2.3.3.1 Definition 12
5.2.3.3.2 Attributes 13
5.2.4 Information attribute definitions 13
5.2.4.1 Introduction 13
5.2.4.2 Definitions and legal values 13
6 Policy MnS – Stage 3 14
6.1 RESTful HTTP-based solution set 14
6.1.1 Mapping of operations 14
6.1.1.1 Introduction 14
6.1.1.2 Operation 14
6.1.2 Mapping of notifications 14
6.1.2.1 Introduction 14
6.1.2.2 Notification 14
6.1.2.2.0 General 14
6.1.2.2.1 Notification "notifyPolicyConflict" 14
6.1.3 Resources 15
6.1.4 Data type definitions 15

6.1.4.1 Query, message body and resource data types 15
6.1.4.1.0 General 15
6.1.4.1.1 Notification "notifyPolicyConflict" 15
6.1.4.2 Referenced structured data types 15
6.1.4.3 Simple data types and enumerations 15
6.1.4.3.0 General 15
6.1.4.3.1 Simple data types 16
6.1.4.3.2 Enumeration notificationType-Type 16
6.2 YANG/Netconf-based solution set 16

Annex A (informative): Change history 17

History 18

Foreword

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In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

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should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

Introduction

The present document is part of a TS-family covering the 3rd Generation Partnership Project Technical Specification Group Services and System Aspects Management and orchestration of networks, as identified below:

TS 28.555: Management and orchestration; Network policy management for 5G mobile networks; Stage 1 [2].

TS 28.556: Management and orchestration; Network policy management for 5G mobile networks; Stage 2 and stage 3 [3].

1 Scope

The present document specifies policy management procedures, stage 2 and stage 3 for policy MnS.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 28.555: "Management and orchestration; Network policy management for 5G mobile networks; Stage 1".

[3] 3GPP TS 28.556: "Management and orchestration; Network policy management for 5G mobile networks; Stage 2 and stage 3".

[4] 3GPP TS 28.532: "Management and orchestration; Generic management services".

[5] 3GPP TS 28.622: "Telecommunication management; Generic Network Resource Model (NRM) Integration Reference Point (IRP); Information Service (IS)".

[6] ITU-T Recommendation X.733 (02/92): "Information technology - Open Systems Interconnection - Systems Management: Alarm reporting function".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

4 Policy management procedures

4.1 Policy Creation

The Figure 4.1-1 illustrates the procedure for creating a new policy.

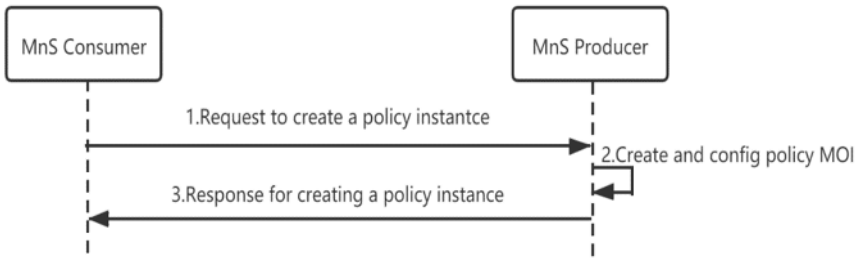


Figure 4.1-1: Procedure for creating a policy

1) MnS Consumer sends a request to create a policy instance to MnS Producer for the new policy to be created(see createMOI operation defined in TS 28.532 [4]). Based on the request, the MnS Producer creates the concrete policy MOI].

2) MnS Producer sends a response to the MnS consumer(see createMOI operation defined in TS 28.532 [4]).

4.2 Policy Deletion

The Figure 4.2-1 illustrates the procedure for deleting a policy.

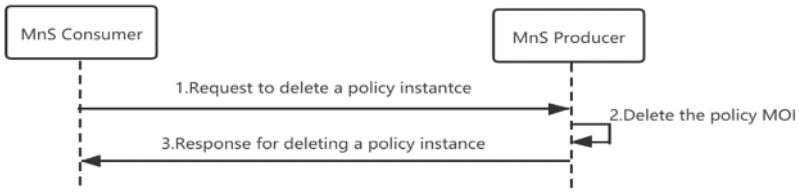


Figure 4.2-1: Procedure for deleting a policy

1) MnS Consumer sends a request to delete a policy instance(see deleteMOI operation defined in TS 28.532 [4]) to MnS Producer.

2) Based on the request, the MnS Producer deletes the concrete policy MOI (i.e. instance of policy IOC).

3) MnS Producer sends a response to the MnS consumer(see deleteMOI operation defined in TS 28.532 [4]).

4.3 Policy Update

The Figure 4.3-1 illustrates the procedure for updating a policy.

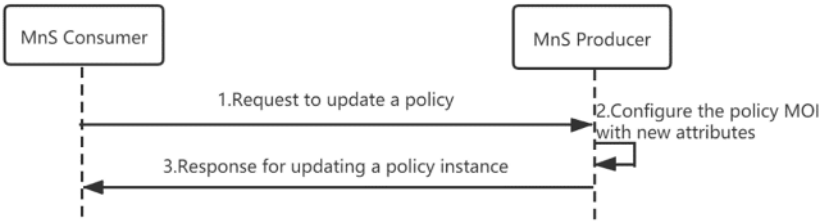


Figure 4.3-1: Procedure for updating a policy

1) MnS Consumer sends a request to update a policy instance to MnS Producer(see modifyMOIAttributes operation defined in TS 28.532 [4]).

2) Based on the request, the MnS Producer updates the concrete policy MOI (i.e. instance of policy IOC).

3) MnS Producer sends a response to the MnS consumer(see modifyMOIAttributes operation defined in TS 28.532 [4]).

4.4 Policy Query

The Figure 4.4-1 illustrates the procedure for querying a new policy.

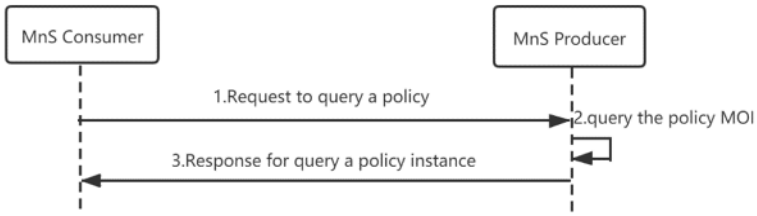


Figure 4.4-1: Procedure for querying a policy

1) MnS Consumer sends a request to query a policy instance to MnS Producer(see getMOIAttributes operation defined in TS 28.532 [4]).

2) Based on the request, the MnS Producer queries the concrete policy MOI (i.e. instance of policy IOC).

3) MnS Producer sends a response to the MnS consumer(see getMOIAttributes operation defined in TS 28.532 [4]).

4.5 Policy Conflicts Notification

The Figure 4.5-1 illustrates the procedure for notifying policy conflicts.

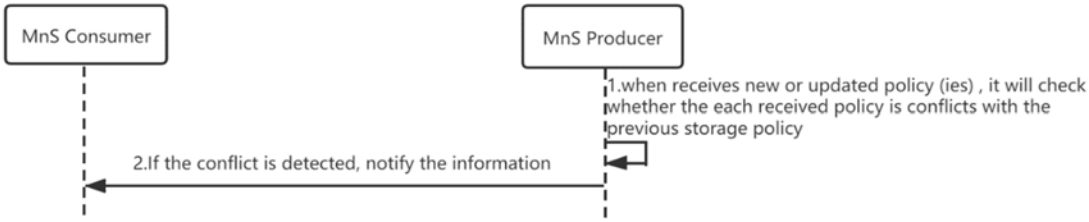


Figure 4.5-1: Procedure for policy conflicts notification

1) 1. When MnS Producer receives new or updated policy (ies) from MnS Consumer, it will check whether the each received policy is conflicts with the previous storage policy.

2) 2. If the conflict is detected, MnS Producer will notify the information to MnS Consumer(see notifyEvent operation defined in TS 28.532 [4]).

4.6 Policy Activation

The Figure 4.6-1 illustrates the procedure for activating a new policy.

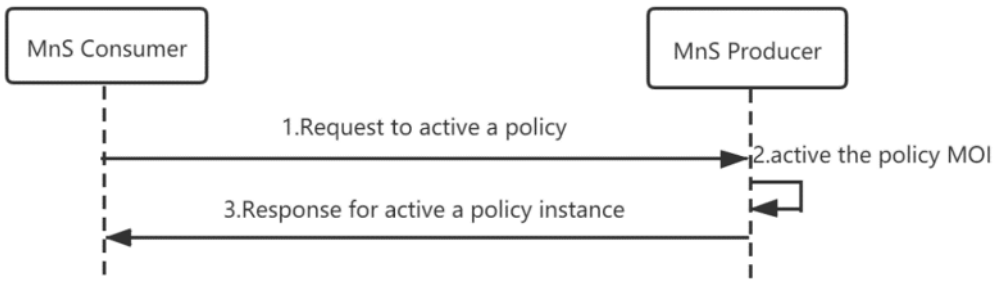


Figure 4.6-1: Procedure for activating a policy

1) MnS Consumer sends a request to activate a policy instance to MnS Producer.
2) Based on the request, the MnS Producer activate the concrete policy MOI (i.e. instance of policy IOC).
3) MnS Producer sends a response to the MnS Consumer.

4.7 Policy Deactivation

The Figure 4.7-1 illustrates the procedure for deactivating a policy.

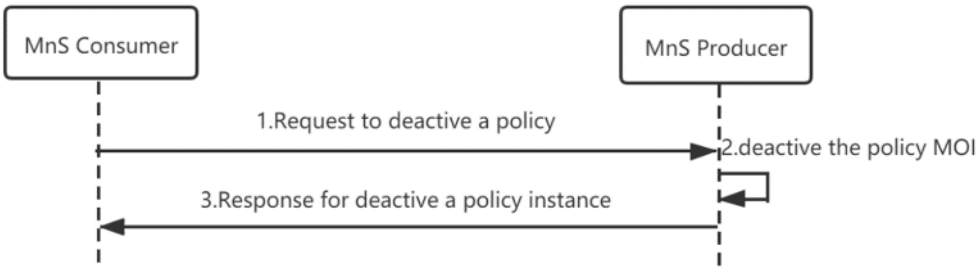


Figure 4.7-1: Procedure for deactivating a policy

1) MnS Consumer sends a request to deactivate a policy instance to MnS Producer.
2) Based on the request, the MnS Producer deactivate the concrete policy MOI (i.e. instance of policy IOC).
3) MnS Producer sends a response to the MnS Consumer.

5 Policy MnS – Stage 2

5.1 Management operation for Policy (MnS component type A)

The operations (e.g. createMOI operations) and notifications (e.g. notifyMOIcreation) of generic provisioning MnS defined in TS 28.532 [4] can be used for policy lifecycle management. The policy can be treated as object instance.

5.2 Information model definition entities for Policy

5.2.1 Imported information entities and local labels

Label reference	Local label
TS 28.622 [5], IOC, `Top`	`Top`
TS 28.622 [5], IOC, `SubNetwork`	`SubNetwork`

5.2.2 Class diagram

5.2.2.1 Relationships

This clause introduces the set of Information Object Classes (IOCs).

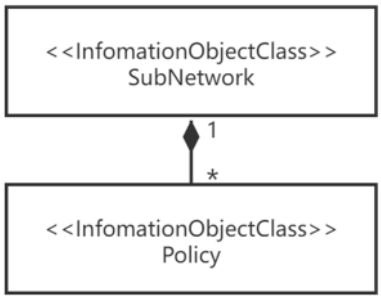


Figure 5.2.2.1: Information Object Class UML Diagram

NOTE: The diagram maybe updated depending on the scenarios.

5.2.2.2 Inheritance

This clause depicts the inheritance relationships.

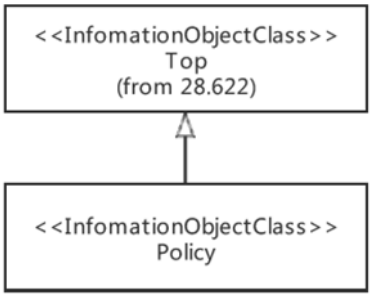


Figure 5.2.2.2: Information Object Class UML Diagram

5.2.3 Class definitions

5.2.3.1 Policy

5.2.3.1.1 Definition

This IOC represents a network policy.

5.2.3.1.2 Attributes

The `Policy` IOC includes attributes inherited from Top IOC (defined in TS 28.622) and the following attributes:

Attribute name	Support Qualifier	isReadble	isWriteble	isInvariant	isNotifyable
`policyPriority`	M	T	T	F	T
`policyStatus`	M	T	T	F	T
`policyType`	M	T	T	F	T
`policyContent`	M	T	T	F	T

5.2.3.2 Notifications

5.2.3.2.0 General

The Notifications include notifications (e.g. notifyMOIcreation) of generic provisioning MnS defined in TS 28.532[4] and Notification `notifyPolicyConflict`(5.2.3.2.1)

5.2.3.2.1 Notification `notifyPolicyConflict` (CM)

5.2.3.2.1.1 Definition

This notification supports the policy conflict notification to be notified when a new policy conflicts with the previous storage policy.

5.2.3.2.1.2 Notification information

Parameter Name	Qualifier	Information Type	Comment
`objectClass`	M, Y	It shall carry the ManagedEntity class name.	It indicates the class, whose instance emitted this notification.
`objectInstance`	M, Y	DN of the instance of the "ManagedElement"	It identifies the instance of the sender of this notification.
`notificationId`	M, N	This is an identifier of the notification, which may be used to correlate notifications.	The unique identifier of the notification across all notifications sent by a particular management service producer throughout the time that correlation is significant. How identifiers of notifications are re-used to correlate notifications is outside of the scope of the present document.
`eventTime`	M, Y	It indicates the event occurrence time.	The semantics of Generalised Time specified by ITU-T shall be used here.
`systemDN`	C	It shall carry the DN of management service producers.	-
`notificationType`	M, Y	"notifyPolicyConflict"	The type of notification, and it shall be assigned to "notifyThresholdCrossing" for this notification.
`conflictDescription`	M, Y	It specifies the policy conflict details.	conflictDetails shall convey more information about the conflict. Description of the detected policy conflicts, e.g. conflicting events, conditions or actions among the policies. Typically, this is due to the fact that a policy is executing, or that Some attributes that require policy execution are missing.

5.2.3.3 PolicyContent <<dataType>>

5.2.3.3.1 Definition

This data type represents the content of a network policy.

5.2.3.3.2 Attributes

Attribute name	Support Qualifier	isReadble	isWriteble	isInvariant	isNotifyable
condition	M	T	T	F	T
action	M	T	T	F	T

5.2.4 Information attribute definitions

5.2.4.1 Introduction

This clause defines the semantics of the attributes used in IOCs.

5.2.4.2 Definitions and legal values

Attribute Name	Definition	Legal Values
policyPriority	It specifies the priority of Policy. allowedValues:LOW,Medium,High	type: ENUM multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: Null isNullable: False
policyStatus	It specifies the status of Policy. If a policy is activated, and then its status is activated. If it is deactivated ,then its status is deactivated allowedValues:ACTIVATED,DEACTIVATED	type: ENUM multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: DEACTIVATED isNullable: False
policyType	It specifies the type of Policy. allowedValues: N/A NOTE: Its value is not defined in the present document.	type: string multiplicity: 0..N isOrdered: N/A isUnique: N/A defaultValue: Null isNullable: False
policyContent	It identifies the content of a network policy A policyContent <<dataType>> condition and action. allowedValues: N/A	type: PolicyContent multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: Null isNullable: False
condition	It identifies the condition of the policy. The type of condition depends on the concrete policy. allowedValues: N/A	type: None multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: Null isNullable: False
action	It identifies the action of policy. The type of action depends on the concrete policy. allowedValues: N/A	type: None multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: Null isNullable: False

6 Policy MnS – Stage 3

6.1 RESTful HTTP-based solution set

6.1.1 Mapping of operations

6.1.1.1 Introduction

The IS operations are mapped to SS equivalents according to table 6.1.1.1-1.

Table 6.1.1.1-1: Mapping of IS operations to SS equivalents

IS operation	HTTP Method	Resource URI	SQ
createMOI	PUT	http://{URI-DN-prefix}/{root}/PoliMnS/{MnSVersion}/ {LDN-first-part}/{className}={id}	M
getMOIAttributes	GET	http://{URI-DN-PREFIX}/{root}/PoliMnS/{MnSVersion}/{LDN-first-part}/{className}={id}	M
modifyMOIAttributes	PUT PATCH	http://{URI-DN-prefix}/{root}/PoliMnS/{MnSVersion}/{LDN-first-part}/{className}={id}	M
deleteMOI	DELETE	http://{URI-DN-prefix}/{root}/PoliMnS/{MnSVersion}/{LDN-first-part}/{className}={id}	M

6.1.1.2 Operation

The mapping of operations (e.g. createMOI operations) of generic provisioning MnS defined in TS 28.532[4] can be used for policy lifecycle management.

6.1.2 Mapping of notifications

6.1.2.1 Introduction

The IS notifications are mapped to SS equivalents according to table 6.1.2.1-1.

Table 6.1.2.1-1: Mapping of IS notifications to SS equivalents

IS notifications	HTTP Method	Resource URI	SQ
`notifyMOICreation`	POST	/notificationSink	M
`notifyMOIDeletion`	POST	/notificationSink	M
`notifyMOIAttributeValueChange`	POST	/notificationSink	M
`notifyMOIChanges`	POST	/notificationSink	M
`notifyPolicyConflict`	POST	/notificationSink	M

6.1.2.2 Notification

6.1.2.2.0 General

The mapping of notifications (e.g. notifyMOICreation) of generic provisioning MnS defined in TS 28.532[4] and notifyPolicyConflict(6.1.1.2.1) can be used for policy lifecycle management.

6.1.2.2.1 Notification "notifyPolicyConflict"

The IS notification parameters are mapped to SS equivalents according to table 6.1.1.2.1-1.

Table 6.1.1.2.1-1: Mapping of IS notification input parameters to SS equivalents (HTTP POST)

IS operation parameter name	SS parameter location	SS parameter name	SS parameter type	SQ
`objectClass objectInstance`	request body	href	uri-Type	M
`notificationId`	request body	notificationId	notificationId-Type	M
`notificationType`	request body	notificationType	notificationTyp-Type	M
`eventTime`	request body	eventTime	dateTime-Type	M
`systemDN`	request body	systemDN	systemDN-Type	M
`conflictDescription`	request body	conflictDescription	conflictDescription-Type	M

6.1.3 Resources

The resources of generic provisioning MnS defined in TS 28.532 [4] can be used for policy lifecycle management.

6.1.4 Data type definitions

6.1.4.1 Query, message body and resource data types

6.1.4.1.0 General

Query, message body and resource data types of generic provisioning MnS defined in TS 28.532 [4] and the following definitions can be used for policy lifecycle management.

6.1.4.1.1 Notification "notifyPolicyConflict"

Table 6.1.4.1.1-1: Definition of type notifyPolicyConflict-NotifType

Attribute name	Data type	Description	SQ
href	uri-Type	URI of the resource where the event (alarm) occurred	M
notificationId	notificationId-Type	Notification identifier as defined in ITU-T Rec. X. 733 [6]	M
notificationType	notificationType-Type	Notification type (notifyPolicyConflict)	M
eventTime	dateTime-Type	Event (PolicyConflict) occurrence time	M
systemDN	systemDN-Type	System DN	M
correlatedNotifications	array(correlatedNotification-Type)	Set of all notifications to which this notification is considered to be correlated as defined in ITU-T Rec. X. 733 [6]	O
additionalText	additionalText-Type	Allows a free form text description to be reported as defined in ITU-T Rec. X. 733 [6]	O
sourceIndicator	sourceIndicator-Type	Indicates the source of the operation that led to the generation of this notification.	O
attributeList	map(anyValue)	The attributes (name/value pairs) of the conflicting MOI.	O

6.1.4.2 Referenced structured data types

The referenced structured data types of generic provisioning MnS defined in TS 28.532 [4] can be used for policy lifecycle management.

6.1.4.3 Simple data types and enumerations

6.1.4.3.0 General

The simple data types and enumerations of generic provisioning MnS defined in TS 28.532 [4], and the following simple data types and enumerations can be used for policy lifecycle management.

6.1.4.3.1 Simple data types

Table 6.1.4.3.1-1: Simple data types

Type name	Type definition	Description
conflictDescription-Type	string	Description of the policy conflict details.

6.1.4.3.2 Enumeration notificationType-Type

Table 6.1.4.3.2-1: Enumeration notificationType-Type

Enumeration value	Description
notifyPolicyConflict	Notification type is notifyPolicyConflict

6.2 YANG/Netconf-based solution set

The YANG/Netconf based solution set of generic provisioning MnS defined in TS 28.532 [4] can be used for policy lifecycle management.

Annex A (informative): Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2020-08	SA5#132e	S5-204442				Skeleton	0.0.0
2020-11	SA5#134e	S5-206030 S5-206365				pCR 28.556 add scope pCR 28.556 add skeleton	0.1.0
2021-05	SA#137e	S5-213015 S5-213017 S5-213018				pCR 28.556 add introduction pCR 28.556 add policy management procedures pCR 28.556 add management operation for Policy	0.2.0
2021-09	SA#138e	S5-214089 S5-214094				pCR 28.556 add policy activation management procedure pCR 28.556 add policy deactivation management procedure	0.3.0
2021-10	SA#139e	S5-215543 S5-215544 S5-215545 S5-215546				pCR 28.556 Add information model definition entities for Policy pCR 28.556 Add class diagram pCR 28.556 Add class definition pCR 28.556 Add Information attribute definitions	0.4.0
2021-11	SA#140	S5-216455 S5-216456 S5-216457 S5-216458 S5-216459				pCR 28.556 Add notification definition pCR 28.556 Change stage 2 information attribute definition pCR 28.556 update Policy activation and deactivation procedure pCR 28.556 Add stage 3 definition pCR 28.556 update PolicyContent definition EditHelp review	0.5.0
2021-12	SA#94e	SP-211419				Presented for information and approval	1.0.0
2021-12	SA#94e					Upgrade to change control version	17.0.0
2023-12	SA#102	SP-231493	0002	1	F	Rel-17 CR 28.556 Correction of attribute properties	17.1.0
2024-04	-	-	-	-	-	Update to Rel-18 version (MCC)	**18.0.0**

History

Document history		
V18.0.0	May 2024	Publication